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Software Requirements Specification

Amazon India E-Commerce Platform — v1.0

Abstract

This Software Requirements Specification (SRS) establishes the formal functional, non-functional, operational, and architectural requirements for the Amazon India E-Commerce Platform in accordance with IEEE 830 standards. Operating as a distributed, high-concurrency marketplace, the platform interconnects consumers, third-party merchants, warehouse fulfillment centers, logistics personnel, and system administrators. The document defines comprehensive capabilities across product discovery, persistent cart synchronization, secure multi-rail financial transactions (UPI, cards, net banking, wallets, COD), real-time order lifecycle tracking, customer support escalation, and merchant catalog governance. Additionally, strict quantitative constraints are enforced regarding sub-300 millisecond response latencies, 99.999% platform availability, cryptographic credential hashing, and statutory data localization compliance.

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1. Introduction

1.1 Purpose

Define the functional, non-functional, security, performance, and operational requirements for the Amazon India E-Commerce Platform. The system supports product discovery, shopping cart management, secure checkout, payments, order management, delivery tracking, customer support, reviews, promotions, and third-party seller operations.

1.2 Scope

A web and mobile e-commerce platform connecting customers, third-party sellers, delivery agents, payment providers, and administrators. The scope includes product search and discovery, shopping cart, checkout, multiple payment methods, order tracking, customer accounts, seller management, reviews and ratings, promotions, customer support, and logistics integration.

1.3 Intended Audience

Software engineers, database developers, UI/UX designers, QA engineers, testers, product managers, system administrators, and project stakeholders.

1.4 Assumptions & Dependencies

Users are assumed to have a stable internet connection and use modern web browsers or supported Android/iOS devices. Payment processing depends on external banking, UPI and payment gateway services, while delivery estimates and tracking depend on external logistics and routing services. The system also depends on cloud infrastructure and third-party APIs.

2. User Roles & Stakeholder Classes

The platform defines distinct stakeholder roles to ensure appropriate authorization boundaries, data isolation, and workflow specialization across the marketplace architecture:

Role	Description & Operational Responsibilities
Shopper / Customer	Browses products, searches and filters items, reads reviews, manages the cart, places orders, makes payments, tracks deliveries, and raises support requests.
Seller	Uploads products and inventory, manages pricing, applies promotions, processes customer orders, and uses the seller dashboard.
Delivery Agent	Scans packages, follows optimized delivery routes, updates delivery status, and verifies successful delivery using OTP or digital confirmation.
Warehouse Staff	Handles product inventory intake, bin allocation, pick-and-pack fulfillment, package labeling, and dispatch to carrier hubs.
Administrator	Monitors the platform, manages categories and site configuration, handles major disputes, and oversees overall system operations.

2.1 Role-Based Access Control Architecture

Access privileges are governed strictly via Role-Based Access Control (RBAC). Customer roles operate within authenticated consumer sessions; merchant accounts require verified GSTIN credentials; logistics agents interface through authenticated hardware-bound mobile terminals; and administrators require multi-factor hardware keys with dedicated VPN tunneling.

3. Functional Requirements

The functional requirements enumerate the core capabilities, operations, and services provided by the Amazon India platform to its stakeholders:

#	Category	Requirement Description	Priority
FR-01	Product Search	The system shall allow customers to search products using keywords, categories, voice search, filters, typo tolerance, and regional-language synonyms.	High
FR-02	Product Discovery	The system shall allow customers to filter products by price, brand, ratings, and delivery eligibility.	Medium
FR-03	Shopping Cart	The system shall allow customers to add, remove, and modify product quantities in their shopping cart.	High
FR-04	Cart Synchronization	The system shall persist and synchronize shopping cart data across web and mobile sessions.	High
FR-05	Checkout	The system shall provide a multi-step checkout process including address selection, delivery options, payment, and order confirmation.	High
FR-06	Payment	The system shall support UPI, credit/debit cards, net banking, Amazon Pay Wallet, and Cash on Delivery.	High
FR-07	Order Management	The system shall generate a unique order ID for every successful transaction and allow customers to view order history.	High
FR-08	Order Tracking	The system shall display order progress through statuses such as Ordered, Processing, Shipped, Out for Delivery, Delivered, and Returned.	High
FR-09	Order Cancellation	The system shall allow customers to cancel an order before it reaches the shipped stage.	Medium
FR-10	Account Management	The system shall allow customers to manage profile information, multiple delivery addresses, and saved payment methods.	Medium
FR-11	Customer Support	The system shall provide chatbot-based customer support and transfer unresolved issues to a human support agent.	Medium
FR-12	Notifications	The system shall send users notifications when important order or delivery milestones occur.	Medium

4. Non-Functional Requirements

Non-functional requirements specify quality attributes, performance benchmarks, architectural constraints, and regulatory standards governing the platform:

#	Category	Requirement Description	Priority
NFR-01	Performance	Search responses shall normally be returned within 300 milliseconds.	High
NFR-02	Performance	Homepage and initial search results shall render within 2 seconds on standard 4G connections.	High
NFR-03	Scalability	The platform shall support extremely high concurrent transaction volumes during major sales events.	High
NFR-04	Security	Passwords shall be securely salted and hashed using a modern password-hashing algorithm such as Argon2.	High
NFR-05	Payment Security	Raw CVV data shall never be stored or logged, and saved card information shall be tokenized.	High
NFR-06	Reliability	The platform shall target 99.999% availability using highly available infrastructure.	High
NFR-07	Disaster Recovery	Core databases shall be backed up at regular intervals and the disaster recovery process shall target less than 30 minutes of maximum tolerable downtime.	Medium
NFR-08	Data Protection	Critical user and payment-related information shall comply with applicable Indian data protection and localization requirements.	High
NFR-09	Accessibility	The interface shall remain responsive across desktop, tablet, and smartphone screen sizes.	Medium

4.1 Statutory Standards & Compliance Mandates

Standard	Compliance Mandate
PCI DSS v4.0	Enforces tokenization for all stored credit/debit card numbers and prohibits CVV retention.
RBI Directives	Mandates Indian data localization for all financial transaction records and UPI tokenization.

5. System Use Cases

Use cases define discrete actor interactions with the system, capturing input stimuli, operational sequences, and system responses.

5.1 Customer Journey Use Cases

UC-1: Search and Browse Products

Actor: Shopper / Customer

Flow: 1. Enter product name or keyword. 2. View search results. 3. Apply filters (price, brand, rating). 4. Select a product.

UC-2: Purchase Product

Actor: Shopper / Customer

Flow: 1. Add product to cart. 2. Proceed to checkout. 3. Select address and delivery option. 4. Make payment. 5. Confirm order.

UC-3: Track Order

Actor: Shopper / Customer

Flow: 1. Open Your Orders. 2. Select an order. 3. Click Track Package. 4. View delivery status.

UC-4: Manage Account

Actor: Shopper / Customer

Flow: 1. Open Account Settings. 2. Add or edit address. 3. Verify PIN code. 4. Save changes.

UC-5: Review Product

Actor: Shopper / Customer

Flow: 1. Open purchased product. 2. Select Write a Review. 3. Give rating and feedback. 4. Submit review.

5.2 Operations & Management Use Cases

Operational workflows governing merchant catalog management, logistics fulfillment, warehouse dispatch, and administrative governance:

UC-6: Manage Product Listings

Actor: Seller

Flow: 1. Open Seller Dashboard. 2. Add or edit product listings. 3. Update inventory and price. 4. Manage active promotions.

UC-7: Manage Products & Orders

Actor: Seller

Flow: 1. Open Seller Dashboard. 2. Add or edit product details. 3. Update inventory and price. 4. View customer orders. 5. Process orders.

UC-8: Manage Deliveries

Actor: Delivery Agent

Flow: 1. View assigned deliveries on mobile app. 2. Scan the package barcode. 3. Follow optimized delivery route. 4. Update transit status. 5. Request delivery OTP from customer.

UC-9: Deliver Order

Actor: Delivery Agent

Flow: 1. View assigned order. 2. Scan the package. 3. Follow the delivery route. 4. Update delivery status. 5. Verify delivery using OTP. 6. Mark order as delivered.

UC-10: Manage Platform

Actor: Administrator

Flow: 1. Open Admin Dashboard. 2. Monitor platform activity. 3. Manage product categories and site settings. 4. Handle major customer-seller disputes. 5. Update platform configurations.

UC-11: Manage Inventory & Shipments

Actor: Warehouse Staff

Flow: 1. View pending orders. 2. Check product availability. 3. Pick and pack products. 4. Update inventory. 5. Generate shipping details. 6. Mark package as dispatched.

6. Behavioral Analysis & System Models

6.1 System Boundary & Actor Interactions

Formal UML use case representation capturing the architectural boundary, external stakeholder roles, and core operational interactions:

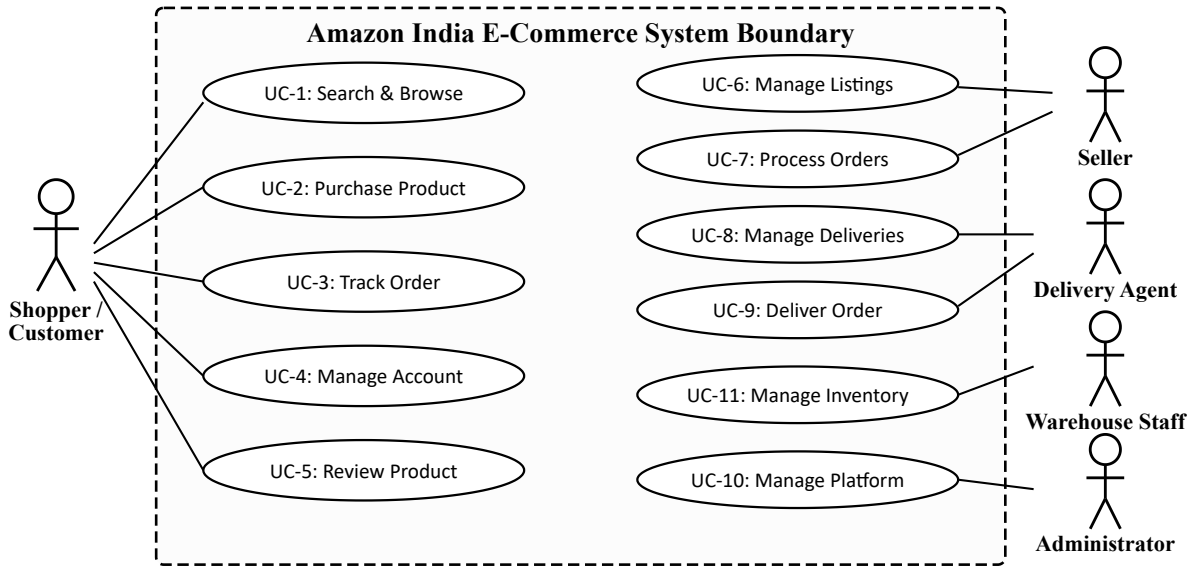


Figure 1: UML Use Case Diagram for Primary Actor Roles

Actor-to-Feature Mapping Summary

Actor Class	Associated Use Cases	Primary System Interface
Customer	UC-1, UC-2, UC-3, UC-4, UC-5	Web Storefront / Consumer Mobile App
Seller	UC-6, UC-7	Amazon Seller Central Dashboard
Delivery Agent	UC-8, UC-9	Logistics Handheld Application
Warehouse Staff	UC-11	Warehouse Management System (WMS)
Administrator	UC-10	Admin Control Center Portal

6.2 Order Checkout & Payment Sequence Model

Sequence interaction diagram illustrating the chronological message exchange, tokenized payment verification, and asynchronous event triggers:

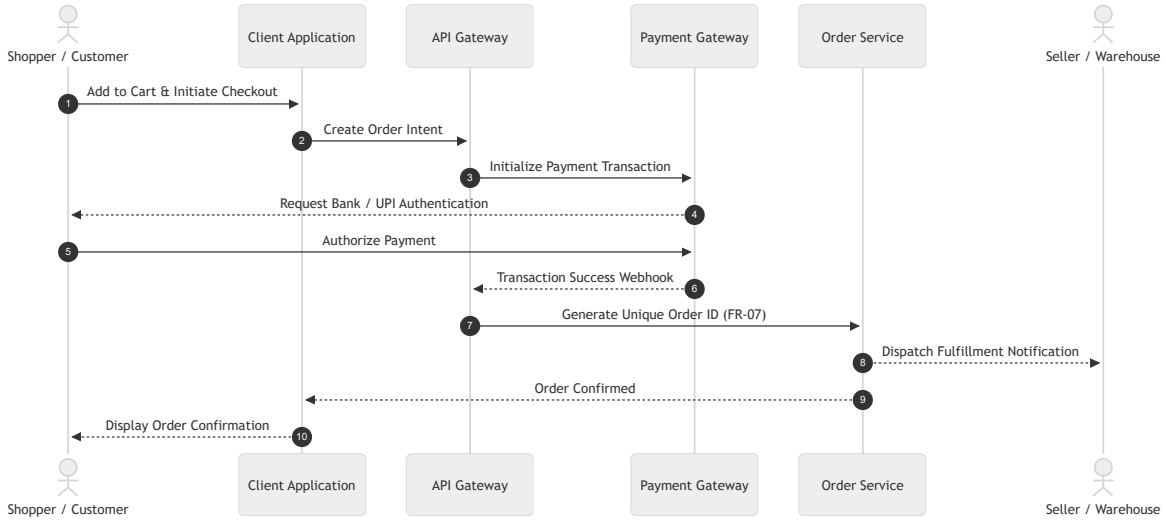


Figure 2: End-to-End Order Checkout Sequence Diagram

Sequence Step Description & Verification

Step #	Interaction Description	Security & Transaction Protocol
1 – 3	Checkout initiation and token creation.	TLS 1.3 encrypted REST payload; zero sensitive card data stored.
4 – 6	Customer authentication and webhook callback.	Two-factor OTP/biometric validation via NPCI/bank payment rail.
7 – 10	Order creation, fulfillment notification, and receipt.	Cryptographically generated unique 17-digit Order ID (FR-07).

6.3 Order Lifecycle State Machine Model

The state machine model specifies the discrete transactional states and event-driven triggers governing the end-to-end lifecycle of consumer orders within the Amazon India marketplace:



Figure 3: Order Lifecycle State Machine Diagram

State Transition Rules & Event Triggers

State	Trigger Event	System Actions & Notifications
Ordered	Successful payment authorization or COD placement.	Generates unique 17-digit Order ID; dispatches SMS/email confirmation.
Processing	Seller acknowledges order; warehouse bin allocation.	Inventory locked; pick-and-pack fulfillment slip generated.
Shipped	Carrier scans parcel barcode at fulfillment hub.	Tracking link activated; courier dispatch milestone notification sent.
Out for Delivery	Agent scans package at last-mile delivery center.	Automated OTP generated and transmitted to customer smartphone.
Delivered	Delivery agent validates OTP at doorstep handover.	Order status marked completed; returns/reviews window opened.
Cancelled	Customer invokes unilateral cancellation prior to dispatch.	Inventory unlocked; automated payment refund workflow initiated.

6.4 IEEE 830 Quality Verification & Project Sign-Off

Formal Quality Assurance (QA) acceptance criteria and engineering verification benchmarks established in accordance with IEEE Std 830-1998 guidelines to validate system compliance before production release:

Verification Phase	Inspection Scope	Acceptance Benchmark
Unit & Integration Testing	Search indexing, cart sync across sessions, payment webhook handling, and inventory deduction.	100% automated test suite pass rate; zero CVV storage validation.
Performance & Stress Testing	Concurrency stress testing simulating seasonal peak load (Great Indian Festival).	Sub-300ms search latency under 50,000 requests/sec peak traffic; <2s initial render.
Security & Regulatory Audit	RBI data localization audit, Argon2 password hash validation, PCI DSS v4.0 tokenization.	Zero critical vulnerabilities; verified database residency within Indian territory.
User Acceptance Testing (UAT)	End-to-end checkout scenarios across web, Android, and iOS device form factors.	Successful execution of all 11 core use cases with verified OTP delivery flows.

Project Sign-Off & Approvals

The undersigned stakeholders certify that this Software Requirements Specification accurately reflects the technical, behavioral, and operational requirements for the Amazon India E-Commerce Platform:

Stakeholder Role	Designation / Authority	Date	Status
Author / Student	Aditya Chandrajit Deore (PRN: 124B1F116)	2026-09-13	Approved
Course Faculty	Department of Computer Engineering, PCCOE	2026-09-13	Verified
Laboratory In-Charge	Software Engineering & Modeling Lab (SEML)	2026-09-13	Completed